SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID:** SIH 26263
- **Problem Statement Title:** Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions
- **Theme:** Smart Automation
- **PS Category:** Software
- **Team ID:** 199751
- **Team Name:** Phoenix Crew

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WAYMARK APAAR HUB

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PROBLEM

- PGDM institutions send detailed student data once a year; others send only total enrolment.
- AICTE cannot easily check who is enrolled, or whether a student is enrolled twice.
- Admissions above approved intake surface late, if at all.
- Employers and regulators have no simple way to verify a student record.

OUR APPROACH

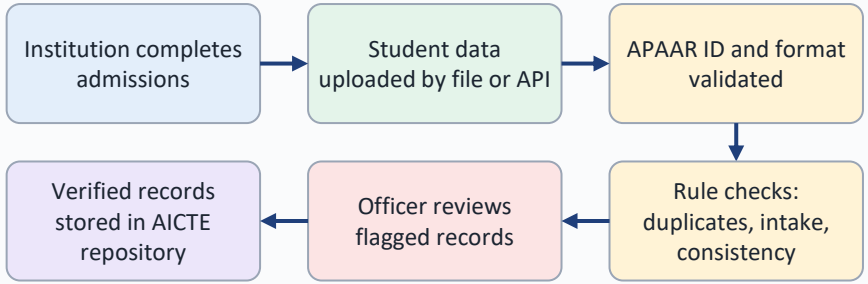
- APAAR ID is the single key for every student record.
- Institutions upload student data after admissions, by file or API.
- Automatic checks catch bad IDs, duplicates and excess intake.
- AICTE monitors live; authorised parties verify with consent.

WHAT IS DIFFERENT

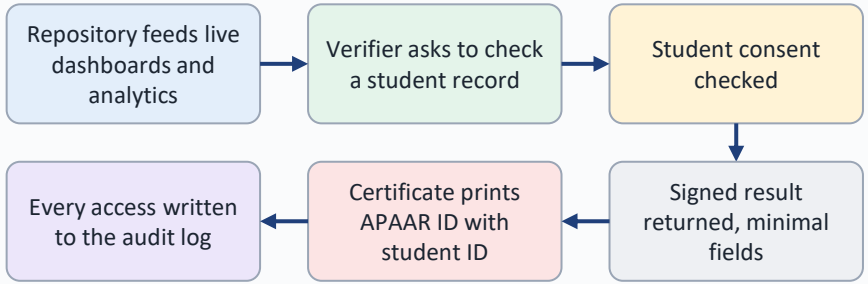
- Every flag shows the rule and the numbers behind it
- Intake check counts supernumerary seats before flagging
- One student at two institutions raises an alert for both
- Audit log chained with SHA-256, so an edited row is caught

USER FLOW

Submission flow



Verification flow



Flags are suggestions. Decisions stay with officers.

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TECHNICAL APPROACH

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INPUT

- Bulk CSV or Excel upload
- REST API for institution systems
- Intake, enrolment and supernumerary seats
- NRI, OCI and FN status, programme details

Proposed: Next.js upload, REST API

VALIDATION

- APAAR ID format and presence
- Duplicates inside and across institutions
- Admissions above intake
- Contradicting fields

Proposed: rules, APAAR lookup adapter

REPOSITORY

- One record per APAAR ID
- Versioned updates with history
- Pending-APAAR status, not rejection
- Hash-chained audit entry per change

Built: FastAPI, SQLAlchemy, audit chain

MONITORING

- Live dashboards for AICTE and institutions
- Intake use by programme
- Rule-based anomaly flags
- TF-IDF match for near-duplicate names

Built: dashboards. Proposed: rules, TF-IDF

OUTPUT

- Verifiable record with QR and signed token
- Reports with CSV export
- Verification API for authorised parties

Proposed: signed token, QR code

CONSENT AND PRIVACY

- Student consent recorded before verification
- Verifier sees only the fields it asks for
- Masking applied in the API, not only the screen

VERIFICATION AND API

- Verifier submits APAAR ID and purpose
- Signed result; APAAR ID printed on certificates
- API and adapters for AICTE and institution systems

SECURITY

- JWT login, bcrypt passwords
- Role checks on every endpoint
- Hash-chained audit log
- HTTPS in transit, encrypted storage

HUMANS DECIDE

- No automatic rejection or penalty
- Each flag shows the rule behind it
- Methods named plainly: rules, TF-IDF

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PLATFORM, VALIDATION AND MONITORING

One APAAR-keyed repository for every approved institution, with checks and monitoring on top.

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BUILT IN THE PROTOTYPE

- JWT login and role-based access guards
- Hash-chained audit log with a verify check
- Institution registry; obligation and alert engine
- Compliance dashboard and notifications

PROPOSED FOR APAAR

- Student upload by file and API
- APAAR validation through a registry adapter
- Intake-cap and cross-institution duplicate checks
- TF-IDF near-duplicate name flag
- Consent-based verification, signed records
- Privacy masking of student fields in the API

Institutions reporting | Records submitted | Valid | Flagged | Excess intake | Duplicate enrolment

DATA PER STUDENT

- APAAR ID (unique key)
- Programme, year, admission date
- Approved intake, actual enrolment
- Supernumerary seats
- NRI category, OCI status
- Foreign national (FN) status
- Consent flag and source

INTAKE CHECK (synthetic)

Flagged: 6 above allowed

Approved intake 120
Supernumerary seats 6
Allowed enrolment 126
Actual enrolment 132
Officer reviews the 6 extra records; nothing is rejected automatically.

Validation rules (each flag gives its reason)

Rule	Catches	Result
ID format	Missing ID, or not 12 digits	Fix and resubmit
Registry lookup	ID not found or details do not match	Held for review
Duplicate in file	Same ID twice in one upload	Flag
Across institutions	Same ID active at two institutions	Alert both
Intake cap	Enrolment above intake plus supernumerary	Flag with count
Near duplicate	Similar name and date of birth, different ID	Suggestion only
Consistency	Contradicting or mandatory fields missing	Flag

Institution dashboard (synthetic sample)

Institution	Intake	Enrolled	Flags	Status
Kaveri Valley College	120	132	3	Excess
Northfield Institute	60	58	1	Alert
Malabar Pharmacy College	40	40	0	Clear

CONSENT, PRIVACY, AUDIT

- Student consent recorded before a verifier sees a record
- Verifier gets only the fields it asks for
- Masking applied in the API, not only the screen
- Every view and change goes to the audit chain

VERIFICATION

- Verifier sends APAAR ID and purpose
- Result: valid or not, signed token and QR
- Certificates print APAAR ID beside student ID
- Roles: AICTE, institution, verifier

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FEASIBILITY, IMPACT AND BENEFITS

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Existing situation vs this platform

Factor	Today (as the PS describes it)	Waymark APAAR Hub
Student data	PGDM: annual detail; others: total enrolment only	Student-level records for all institutions
Identity key	No common student key	APAAR ID on every record
Excess enrolment	Not visible beyond totals	Counted per programme, with reasons
Duplicate enrolment	Not visible across institutions	Flagged for both institutions
Verification	No central route	Consent-based, signed result

WHY IT IS FEASIBLE

- Open-source: Next.js, FastAPI, SQL
- Access, audit chain and alerts already built
- Rules and TF-IDF run without a GPU
- File upload works with no integration

PRACTICAL IMPLEMENTATION

- APAAR lookup via adapter, once authorised
- REST API and file upload for institutions
- AICTE portal integration via adapters
- PostgreSQL, queue, object storage at scale

Potential challenges	How we handle them
Student has no APAAR ID yet	Held as pending APAAR, not rejected
Lookalike mistaken for a duplicate	A flag is a suggestion; an officer decides
Student data is sensitive	Consent, minimal fields, masking, audit
Older institution systems	CSV template and API both supported

Impact and benefits

Feature	Impact	Benefit
APAAR-keyed records	One identity per student	Authentic, verifiable records
Automated validation	Errors caught at upload	Cleaner data, less rework
Intake check	Excess enrolment shown with a count	Early detection
Cross-institution check	Duplicate enrolment alerts	One student, one seat
Live dashboards	AICTE and institutions see the same numbers	Real-time monitoring
Verification	Consent-based, signed result	Trusted checks by employers
Reports	Policy, accreditation, regulatory use	Data-driven governance
Audit and consent	Hash-chained log of every change	Traceable access

Expected beneficiaries

AICTE	**Institutions**	**Students**	**Employers**	**Regulators**
authentic, current student data	one upload, clear errors	one identity, consent control	quick trusted checks	reports for policy

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RESEARCH AND REFERENCES

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1. APAAR Brochure, CIET-NCERT (Ministry of Education)
Relevance: APAAR as a 12-digit national academic identity.

2. Digital Personal Data Protection Act, 2023
Act No. 22 of 2023, Gazette of India, 11 August 2023. Relevance: consent and data principal rights.

3. Anomaly Detection: A Survey (2009)
Chandola, Banerjee, Kumar. ACM Computing Surveys 41(3), Article 15. Relevance: rule and statistical anomaly flags.

4. Lightweight String Similarity Approaches for Duplicate Detection (2025)
Journal of Informatics and Web Engineering, Vol. 4, Issue 3. Relevance: matching near-duplicate names.

5. National Education Policy 2020 (Ministry of Education)
Relevance: policy basis for the Academic Bank of Credits and a national student identity.

PROJECT LINKS

Source code
github.com/Karunesh-18/SIH-AICTE-APAAR

Prototype images
Screenshots link (add before submitting)

Demo video
Walkthrough link (add before submitting)

OFFICIAL REFERENCE

AICTE Approval Process Handbook 2024-2027
Relevance: approved intake and approval norms. The intake and seat figures in this deck are synthetic.

The prototype uses invented data and is not an official AICTE system.